Handling the Ambiguous SSA Case Completely

Answer Key

Precalculus

Quick Answers

1. (a) height h, in cm = 7.71 cm, —
2. 30.44 degrees
3. (a) the acute value of B = 62.11 degrees, (b) the obtuse value of B = 117.89 degrees
4. (a) height h, in cm = 4 cm, —
5. (a) the acute value of B = 69.44 degrees, (b) the obtuse value of B = 110.56 degrees
6. 22.21 cm
7. (a) height h, in cm = 8.6 cm, —
8. (a) the acute bearing angle = 60.25 degrees, (b) the obtuse bearing angle = 119.75 degrees
9. 617.86 m
10. (a) the longer possible c = 70.90 m, (b) the shorter possible c = 18.28 m, —

What is verified

14 of 18 answers machine-checked against SymPy, across 10 problems.
— marks an answer only you can judge — problems 1, 4, 7, 10. The worked solution states what a correct response must contain.

Curriculum

Level: Precalculus
HSG-SRT.D.11 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*
Difficulty 1–5 of 5 across 18 tagged checks.

1. $b = 12$ cm, $a = 6$ cm, $A = 40°$. (a) Find h. (b) How many triangles?

(a) The perpendicular from C to the base line is the leg opposite A in a right triangle whose hypotenuse is the 12 cm side: $h = 12 \sin 40° = 7.7134\ldots$ $\boxed{h \approx 7.71 \text{ cm}}$

(b) *Instructor-judged.* A correct response compares $a = 6$ cm with $h \approx 7.71$ cm and concludes **no triangle**: the side swung down from C is shorter than the perpendicular distance to the base line, so it cannot reach that line at all. Answering "none" without the comparison, or comparing against b instead of h, is not full credit.

2. $a = 14$ cm, $b = 9$ cm, $A = 52°$. Find B.

Solution: $a \geq b$, so the height test is already settled — exactly one triangle, and the angle opposite the shorter side must be acute.

$$\sin B = \frac{9 \sin 52°}{14} = \frac{7.0921\ldots}{14} = 0.5065\ldots, \qquad B = 30.4362\ldots°$$

The obtuse partner $149.56°$ is rejected: with $A = 52°$ it would exceed $180°$. $\boxed{B \approx 30.44°}$

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3. $a = 8$ cm, $b = 10$ cm, $A = 45°$. (a) Acute B. (b) Obtuse B.
Check first: $h = 10 \sin 45° = 7.07$, and $7.07 < 8 < 10$, so $h < a < b$ — two triangles.
(a)

$$\sin B = \frac{10 \sin 45°}{8} = 0.8838\ldots, \qquad B = 62.1144\ldots°$$

$\boxed{B_1 \approx 62.11°}$

(b) The obtuse partner has the same sine: $B_2 = 180° - 62.1144\ldots° = 117.8855\ldots°$. It survives because $45° + 117.89° = 162.89° < 180°$. $\boxed{B_2 \approx 117.89°}$

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4. $b = 8$ cm, $a = 5$ cm, $A = 30°$. (a) Find h. (b) How many triangles?
(a) $h = 8 \sin 30° = 8\left(\frac{1}{2}\right) = 4$. $\boxed{h = 4 \text{ cm}}$
(b) *Instructor-judged.* A correct response places $a = 5$ cm between $h = 4$ cm and $b = 8$ cm and concludes **two triangles**: a side longer than the perpendicular but shorter than the swinging side meets the base line twice, once on each side of the foot of the perpendicular. Stating "two" without that double-intersection reason is partial credit.

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5. $a = 9$ cm, $b = 11$ cm, $A = 50°$. (a) Acute B. (b) Obtuse B.
Check first: $h = 11 \sin 50° = 8.43$, and $8.43 < 9 < 11$ — two triangles.
(a)

$$\sin B = \frac{11 \sin 50°}{9} = 0.9362\ldots, \qquad B = 69.4353\ldots°$$

$\boxed{B_1 \approx 69.44°}$

(b) $B_2 = 180° - 69.4353\ldots° = 110.5646\ldots°$, and $50° + 110.56° < 180°$, so it is a genuine second triangle. $\boxed{B_2 \approx 110.56°}$

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6. $a = 20$ cm, $b = 13$ cm, $A = 63°$. Find c.
Solution: $a > b$, so there is one triangle. Find B, then C, then c.

$$\sin B = \frac{13 \sin 63°}{20} = 0.5790\ldots, \qquad B = 35.3859\ldots°$$

$$C = 180° - 63° - 35.3859\ldots° = 81.6141\ldots°$$

$$c = \frac{20 \sin 81.6141\ldots°}{\sin 63°} = 22.2062\ldots$$

$\boxed{c \approx 22.21 \text{ cm}}$

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7. $b = 15$ cm, $a = 7$ cm, $A = 35°$. (a) Find h. (b) How many triangles?
(a) $h = 15 \sin 35° = 8.6036\ldots$ $\boxed{h \approx 8.60 \text{ cm}}$

(b) *Instructor-judged.* A correct response compares $a = 7$ cm with $h \approx 8.60$ cm: the swinging side is shorter than the perpendicular, so it never reaches the base line and **no triangle** exists. A strong answer adds the algebraic tell — the law of sines would ask for $\sin^{-1}$ of $15 \sin 35°/7 \approx 1.23$, and no angle has a sine above 1. Answering "none" without a comparison is partial credit.

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8. Coastguard: $b = 340$ m, $a = 300$ m, $A = 50°$. (a) Acute B. (b) Obtuse B.

Check first: $h = 340 \sin 50° = 260.5$ m, and $260.5 < 300 < 340$ — two positions are consistent with the measurements.

(a)

$$\sin B = \frac{340 \sin 50°}{300} = 0.8683\ldots, \qquad B = 60.2510\ldots°$$

$\boxed{B_1 \approx 60.25°}$

(b) $B_2 = 180° - 60.2510\ldots° = 119.7490\ldots°$; with $A = 50°$ the angles total $169.75°$, so this triangle exists too. The station would need one more measurement to tell the two apart. $\boxed{B_2 \approx 119.75°}$

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9. Surveyor: $b = 380$ m, $a = 520$ m, $A = 57°$. Find c.

Solution: $a > b$, so exactly one triangle — no second case to carry.

$$\sin B = \frac{380 \sin 57°}{520} = 0.6128\ldots, \quad B = 37.7899\ldots°, \quad C = 85.2101\ldots°$$

$$c = \frac{520 \sin 85.2101\ldots°}{\sin 57°} = 617.8565\ldots$$

$\boxed{c \approx 617.86 \text{ m}}$

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10. Cables: $b = 60$ m, $a = 48$ m, $A = 42°$. (a) Longer c. (b) Shorter c. (c) Explain.

Check first: $h = 60 \sin 42° = 40.15$ m, and $40.15 < 48 < 60$ — two triangles, so two anchor positions.

(a) Projecting onto the ground line, the two possible distances are

$$c = 60 \cos 42° \pm \sqrt{48^2 - (60 \sin 42°)^2} = 44.5904\ldots \pm 26.3070\ldots$$

Taking the plus sign gives the far anchor. $\boxed{c_1 \approx 70.90 \text{ m}}$

(b) Taking the minus sign gives the near anchor. $\boxed{c_2 \approx 18.28 \text{ m}}$

(c) *Instructor-judged.* A correct response says the 48 m cable, swung down from the top of the pole, crosses the ground line twice — once beyond the foot of the perpendicular and once short of it — because it is longer than the 40.15 m perpendicular but shorter than the 60 m stay. The far crossing is the 70.90 m anchor and the near one the 18.28 m anchor; both are genuine, and a complete answer reports both. Choosing one silently, or discarding the shorter distance without a reason, is not full credit.